

TECHNICAL NOTE

PWM Duty Cycle and Digital Resolution

From ideal pulse-width modulation to timer quantization, practical pulse limits,
and closed-loop control behavior

*A practical study of timer quantization, PWM resolution and hardware pulse limits in digitally controlled
power converters*

Abstract

Pulse-width modulation converts a digital timing command into an average electrical action. Although duty cycle is often introduced as a continuous ratio between zero and one, a real digital PWM generator cannot produce an infinite number of duty values. It uses a finite-frequency timer, integer period and compare registers, output logic, and a physical switching stage. Consequently, a controller may calculate a precise floating-point duty command while the PWM peripheral produces a nearby quantized value and the power stage produces a pulse that differs again because of deadtime, propagation delay, switching transitions, and minimum-pulse constraints. This paper develops the subject from first principles. It defines duty cycle and switching frequency, derives timer resolution for edge-aligned and center-aligned PWM, explains quantization error and effective bit resolution, and connects duty resolution to voltage resolution. It then examines the interaction of PWM quantization with proportional and integral control, including steady-state error and limit cycles. The final sections provide a practical method for deciding whether a required duty and correction are physically realizable and for verifying that decision through register inspection and oscilloscope measurement.

1. Introduction

PWM is one of the main interfaces between a digital controller and a power converter. The controller calculates a desired action, such as voltage, current, torque, or power. The PWM peripheral converts that numerical command into switching instants. Semiconductor devices then apply those switching states to an electrical plant. These three layers should not be treated as identical. The numerical command is continuous within the limits of floating-point arithmetic, the timer output is discrete in time, and the practical electrical pulse is shaped by real hardware.

A basic understanding of duty cycle is not enough when the required correction becomes small. In high-voltage or high-frequency systems, one ordinary timer count may represent a meaningful voltage change. A control correction that appears valid in software may be smaller than one timer count and therefore produce no immediate PWM change. Integral action can accumulate the error until the command crosses a count boundary, after which the applied duty changes abruptly. This behavior can contribute to low-amplitude oscillation, current-sharing limit cycles, and differences between software commands and measured waveforms.

2. Duty Cycle and Switching Period

2.1 Definition of duty cycle

A PWM waveform alternates between an ON state and an OFF state. One complete repetition is the switching period. Duty cycle is the fraction of that period spent in the ON state.

$$D = \frac{T_{\text{ON}}}{T_{\text{PWM}}}$$

A duty of zero corresponds to an output that never enters the ON state, while a duty of one corresponds to an output that remains ON for the full period. Duty is often shown as a percentage, so $D = 0.30$ is equivalent to 30%. Duty describes the proportion of a period, not its absolute duration. A 30% duty waveform at 10 kHz has a longer pulse than a 30% duty waveform at 100 kHz.

2.2 Switching frequency

Switching frequency is the number of PWM periods produced each second. It is the reciprocal of the switching period.

$$f_{\text{PWM}} = \frac{1}{T_{\text{PWM}}}$$

At 20 kHz, the period is 50 microseconds. A 30% command therefore produces an ideal ON time of 15 microseconds. At 40 kHz, the period is 25 microseconds and the same duty produces 7.5 microseconds. This distinction matters because timer resolution, deadtime, switching transitions, and minimum-pulse limits are all time-based phenomena.

PWM frequency	Period	ON time at 10%	ON time at 50%
10 kHz	100 us	10 us	50 us
20 kHz	50 us	5 us	25 us
40 kHz	25 us	2.5 us	12.5 us
100 kHz	10 us	1 us	5 us

3. Duty Cycle as an Average Electrical Command

In an ideal switching node connected alternately to a DC source and zero volts, the average node voltage over one period is proportional to duty. If the source voltage is V_{DC} , the ideal average is D multiplied by V_{DC} . An inductor and capacitor may filter the switching waveform so that the load responds mainly to this average rather than to each individual transition.

$$V_{\text{avg}} = DV_{\text{DC}}$$

For an ideal buck converter operating in continuous conduction, this leads to the familiar approximation $V_{\text{out}} = DV_{\text{in}}$ [1]. It is an average model, not a statement that the instantaneous switch-node voltage equals $D V_{\text{in}}$. The switch node still moves between discrete electrical states. Device voltage drops, winding resistance, deadtime, ripple, and discontinuous conduction cause the practical relationship to depart from the ideal expression.

The equation is nevertheless useful because it converts duty resolution into an approximate voltage resolution. If the minimum duty change is ΔD , the corresponding ideal average-voltage step is $V_{\text{in}} \Delta D$. The same PWM timer therefore produces a larger electrical step when the DC-link voltage is higher.

The voltage-resolution relationship used in this note is based on an ideal buck-type conversion stage. Other converter topologies have different duty-to-output relationships.

$$\Delta V_{\text{average}} \approx V_{\text{in}} \Delta D$$

4. Digital Generation of PWM

4.1 Timer clock and time grid

A digital PWM peripheral uses a timer clock. If the timer clock frequency is f_{timer} , one timer tick lasts $1/f_{\text{timer}}$ seconds. Without a high-resolution extension, PWM edges can normally be placed only on this timing grid. A 100 MHz timer has a 10 ns tick; a 200 MHz timer has a 5 ns tick. The controller may calculate an edge time between timer ticks, but an ordinary compare unit must select one of the available timer positions [2].

$$T_{\text{tick}} = \frac{1}{f_{\text{timer}}}$$

4.2 Period and compare registers

The period register defines how many counts form the PWM carrier, while a compare register defines the count at which an output action occurs. Actual endpoint behavior depends on the PWM peripheral, counting mode, compare action and output polarity [2], [6]. In a simple edge-aligned implementation, a period of 5000 counts and a compare value of 1500 produce approximately 30% duty. Changing the compare value from 1500 to 1501 changes the duty by one count. Because the compare register is integer-based, a calculated command such as 1500.65 counts must be rounded or truncated.

$$D_{\text{actual}} \approx \frac{N_{\text{CMP}}}{N_{\text{period}}}$$

Figure 1 illustrates how an integer compare register converts a continuous duty request into a discrete timer edge position.

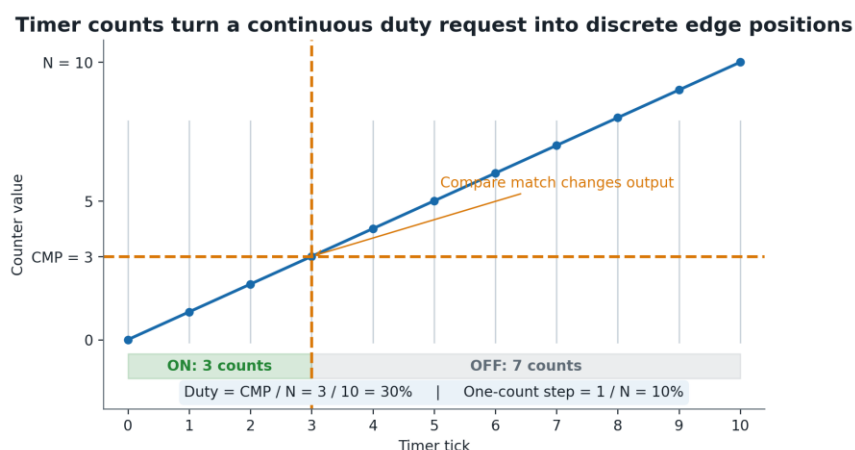


Figure 1. Conversion of a continuous duty request into discrete timer-count positions.

5. Edge-Aligned and Center-Aligned PWM

5.1 Edge-aligned operation

In edge-aligned PWM, the counter normally advances from zero to the period value and then returns to zero. If N timer counts are available in one PWM period, the ordinary duty increment is approximately $1/N$. Ignoring small implementation details concerning whether zero and the terminal count are included, N is approximately the timer-clock frequency divided by the PWM frequency.

$$N_{\text{edge}} \approx \frac{f_{\text{timer}}}{f_{\text{PWM}}}$$

$$\Delta D_{\text{edge}} \approx \frac{1}{N_{\text{edge}}}$$

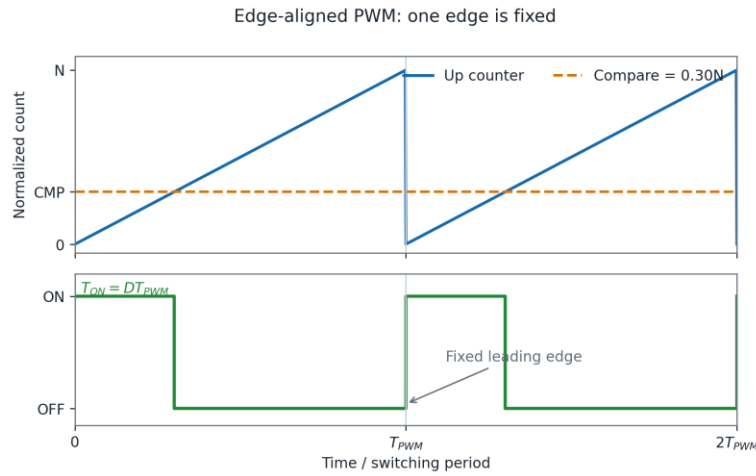


Figure 2. Edge-aligned PWM operation with a fixed leading edge and compare-controlled trailing edge.

5.2 Center-aligned operation

In center-aligned PWM, the counter moves upward to the period value and then downward to zero. A full carrier cycle therefore requires approximately twice the period-register count. The period-register value is consequently about f_{timer} divided by twice the PWM frequency. A one-count compare change produces a duty increment of approximately $1/N_{\text{center}}$.

$$N_{\text{center}} \approx \frac{f_{\text{timer}}}{2f_{\text{PWM}}}$$

$$\Delta D_{\text{center}} \approx \frac{1}{N_{\text{center}}}$$

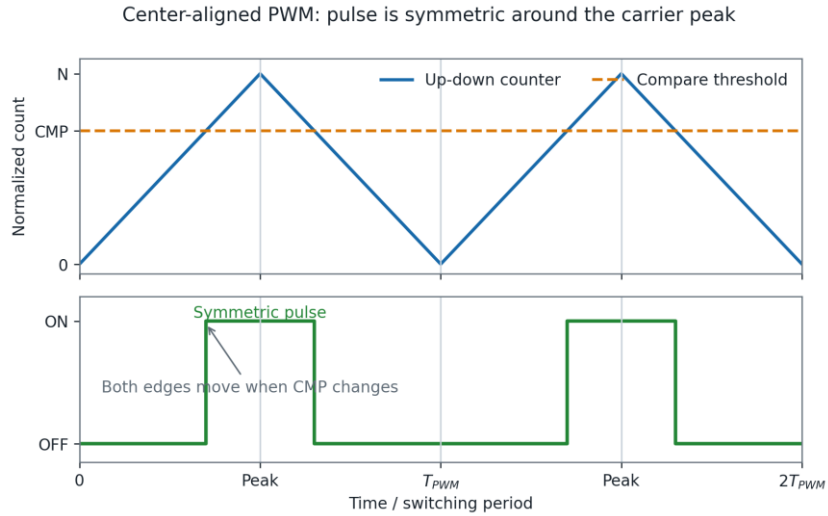


Figure 3. Center-aligned PWM operation with symmetric switching edges.

In the active-high peak-centered example shown in Figure 3, the compare threshold is $N_{CMP} = (1 - D)N_{center}$. The exact compare-to-duty relationship depends on the configured output actions and polarity [2], [6].

At the same timer clock and switching frequency, center-aligned PWM usually provides about half as many ordinary compare levels as edge-aligned PWM. Center alignment remains common in power electronics because it gives symmetrical pulses, useful harmonic properties, and convenient peak-and-valley sampling points. Resolution is one design factor, not the only factor.

6. Frequency and Resolution

Lowering PWM frequency increases the time available in each switching period. Because the timer clock has not changed, a longer period contains more timer counts. More counts provide more possible compare values and therefore a smaller duty increment. Conversely, increasing switching frequency shortens the period, reduces the number of counts, and makes each ordinary duty step larger.

Condition	Timer counts per period	Duty resolution	Other effect
Lower PWM frequency	More	Finer	More current and voltage ripple
Higher PWM frequency	Fewer	Coarser	Less ripple but more switching loss
Higher timer clock	More	Finer	May be limited by the PWM peripheral
High-resolution PWM	Fractional edge placement	Much finer	Needs supported hardware and setup

For example, consider a 100 MHz timer using center-aligned PWM. At 20 kHz, the period register is approximately 2500 counts and one duty step is 0.04%. At 10 kHz, it becomes approximately 5000 counts and one step is 0.02%. Lowering frequency by half doubles the count range and halves the ordinary duty increment.

PWM frequency	Center-aligned period count	Duty step	Effective levels
100 kHz	500	0.20%	about 500
50 kHz	1000	0.10%	about 1000
20 kHz	2500	0.04%	about 2500
10 kHz	5000	0.02%	about 5000

7. Resolution in Counts and Bits

PWM resolution may be stated as a duty increment, a number of levels, or an effective number of bits. If N useful compare levels exist, the approximate effective bit resolution is log base two of N . A peripheral may contain a 16-bit register, but the effective resolution is not 16 bits unless the configured period actually uses close to 65,536 levels.

$$n_{\text{effective}} \approx \log_2(N_{\text{levels}})$$

Usable levels	Effective bits	Approximate duty step
256	8.00	0.391%
1024	10.00	0.0977%
2500	11.29	0.0400%
4096	12.00	0.0244%
65536	16.00	0.00153%

8. Quantization of the Requested Duty

A software controller may calculate duty with many decimal places, but the timer command is obtained by multiplying duty by the period count and converting the result to an integer. If a 5000-count PWM receives a requested duty of 30.013%, the ideal compare value is 1500.65. Rounding gives 1501 counts and an actual duty of 30.020%; truncation gives 1500 counts and an actual duty of 30.000%. The difference between the requested and applied values is quantization error.

$$N_{\text{CMP,ideal}} = D_{\text{requested}}N$$

$$D_{\text{actual}} = \frac{N_{\text{CMP}}}{N}$$

With rounding, the instantaneous quantization error is normally bounded by about half of one least-significant duty step. With truncation, it can approach one full step and may introduce a small bias. Shadow-register loading adds another issue: the compare value may be written during one control calculation but take effect only at the next configured carrier event.

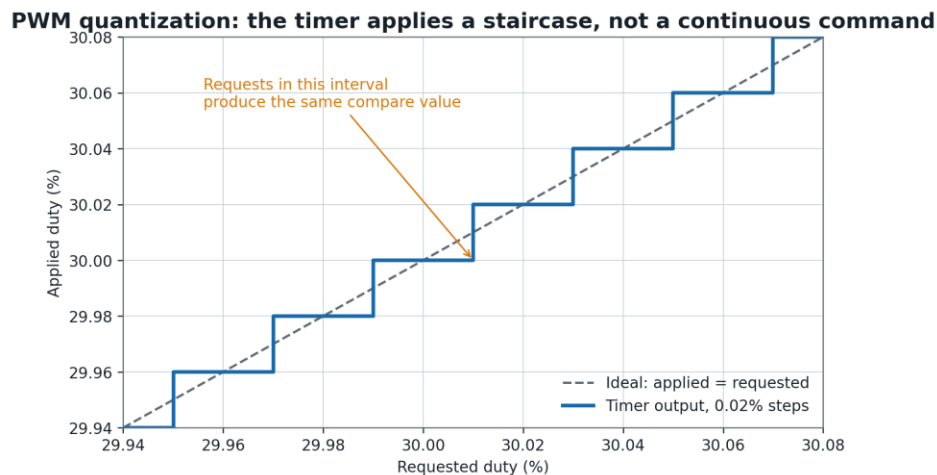


Figure 4. Quantization of requested duty into discrete applied PWM levels.

9. Base Duty and Control Compensation

Closed-loop systems often contain a large base command and a much smaller correction. The final command can be written as $D = D_{\text{base}} + \Delta D$. A base duty of 30% may be easy to generate, while a correction of 0.01% may be smaller than one timer count. It is therefore incorrect to conclude that sufficient base-duty capability guarantees sufficient control resolution.

$$D_{\text{command}} = D_{\text{base}} + \Delta D_{\text{control}}$$

Suppose the minimum PWM step is 0.04%. A correction request of 0.015% cannot be represented as a single fixed compare-count change. The software variable changes, but the physical PWM can remain on the same level. Only when the accumulated command crosses a rounding boundary does the compare register move to the next count. This is why duty variables should be compared with the actual period and compare registers.

10. Interaction with Proportional and Integral Control

10.1 Proportional control

A proportional controller produces an output proportional to error. When the error becomes small, its requested correction may fall below one PWM step. The applied PWM then stops changing even though the error is not zero. The result is a practical steady-state error caused by actuator quantization. Increasing proportional gain can make a small error produce a larger correction, but excessive gain can increase noise sensitivity and oscillation.

$$u_p[k] = K_P e[k]$$

10.2 Integral control

An integral controller accumulates error over time. Even if each individual increment is smaller than one PWM count, repeated increments eventually move the requested duty across a count boundary. The applied duty then changes by one full step. If that step carries the plant past the target, the error reverses and the integral moves back. The resulting repeated motion between adjacent duty levels is a quantization-induced limit cycle.

$$u_i[k] = u_i[k-1] + K_I T_s e[k]$$

Integral action is therefore able to overcome static quantization, but it cannot create a truly fractional fixed compare count. It may instead alternate between neighboring counts so that the long-term average

lies between them. This behavior resembles dithering. It can be acceptable when the plant filters the resulting ripple, but it can also appear as current swapping, audible tones, or low-frequency oscillation.

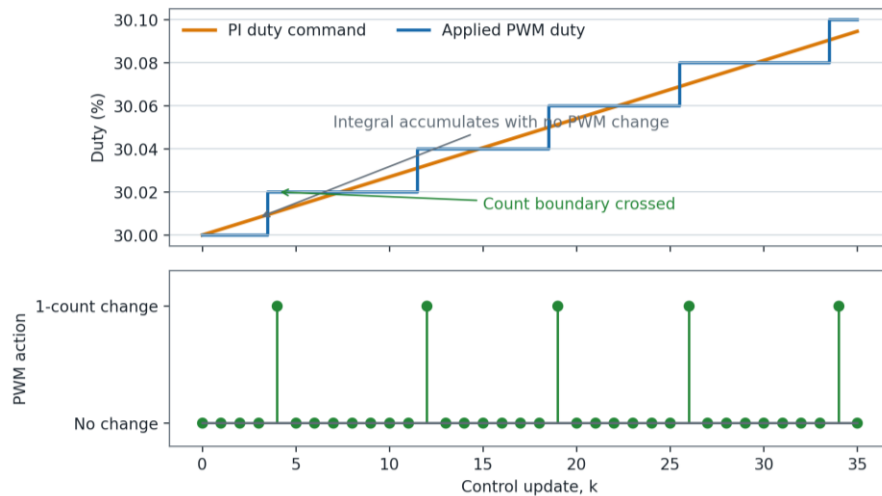


Figure 5. Accumulation of an integral command below one PWM count and the resulting count transitions.

The count-boundary behavior is illustrated in Figure 5.

10.3 Anti-windup and limits

If the requested output cannot be applied because of duty saturation, minimum-pulse suppression, or another actuator limit, an unrestricted integral can continue accumulating. This is windup. When the actuator becomes available again, the stored integral may produce a large transient. Proper designs normally use command limits and anti-windup logic. Removing limits can be useful for a controlled simulation diagnosis, but it does not remove the physical limits of the timer or switching stage.

11. Practical Pulse Limits

11.1 Deadtime

Complementary half-bridge switches require deadtime to prevent simultaneous conduction. During deadtime, both commanded devices are off and current commutates through a diode or channel determined by current direction. Deadtime therefore alters the effective average voltage. Its relative importance grows as the requested pulse becomes shorter. A 500 ns deadtime is only 2% of a 25 microsecond period, but it is 50% of a 1 microsecond pulse.

11.2 Minimum pulse width

The timer may mathematically generate a very short pulse, yet the gate driver and semiconductor may not reproduce it reliably. Practical narrow-pulse behavior depends on gate-driver propagation delay, rise and fall times, loading conditions and circuit parasitic. Propagation delays, rise and fall times, digital isolator delays, pulse filters, desaturation blanking, and driver minimum-pulse specifications all contribute to a practical minimum [4]. Some PWM libraries intentionally suppress pulses below a threshold. Thus, practical resolution must be evaluated in both counts and time. Figure 6 compares ideal timer commands with pulses affected by deadtime and practical minimum-pulse limits.

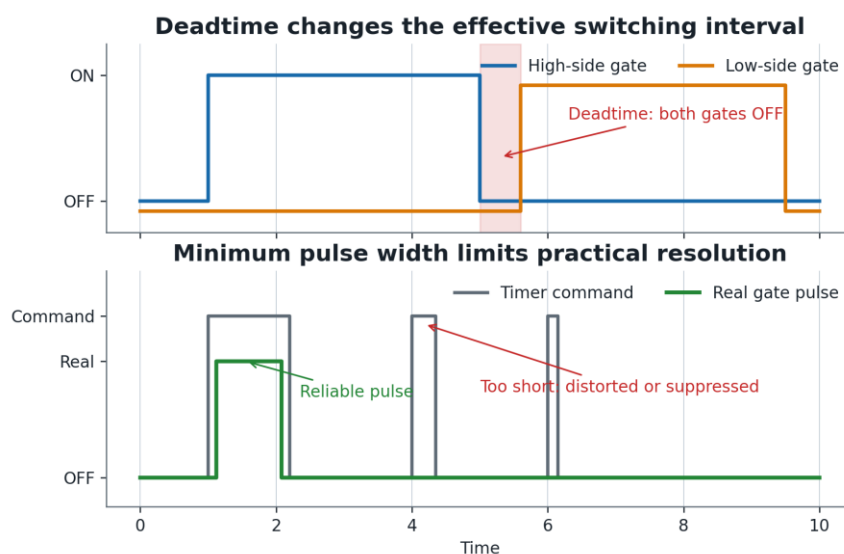


Figure 6. Effects of deadtime and minimum pulse width on the realized gate signal.

11.3 Simulation and HIL time steps

A real-time simulator also has a discrete execution step. A gate pulse that is comparable with one simulation step may be delayed, shortened, or missed according to the input-capture method. The practical HIL limit should therefore include the model step, digital-input sampling rate, and any sub-step switching algorithm [5]. A controller can have adequate hardware resolution while the HIL interface remains unable to reproduce the same edge placement.

12. High-Resolution PWM

High-resolution PWM extends ordinary timer resolution by placing an edge at a fractional position within one timer tick. Dedicated delay hardware may divide a normal tick into many smaller steps [3]. This allows fine duty adjustments without lowering switching frequency. High-resolution operation is not automatic: it may require calibration, supported pins, special compare registers, specific counting modes, and library functions that preserve the fractional command. A device specification stating that HRPWM exists does not prove that a given application enables it.

13. Engineering Evaluation Method

A sound evaluation begins with the required operating range. Determine the minimum and maximum base duty from the converter topology and input-output conditions. Convert each important operating duty into ON time and compare it with deadtime and the practical minimum pulse. Next, determine the smallest correction the controller must apply to meet the required voltage, current, torque, or sharing accuracy.

The PWM configuration must then be reconstructed from actual hardware settings. Identify the source clock, PWM clock dividers, time-base clock, count mode, period register, compare registers, shadow-load event, deadtime count, and any high-resolution configuration. Datasheet maximum values alone are insufficient because software dividers and library settings determine the active configuration.

$$\Delta N_{\text{required}} = \frac{\Delta D_{\text{required}}}{\Delta D_{\text{timer}}}$$

A correction below one count cannot be represented as a constant ordinary PWM level. A correction of several counts should be clearly realizable. Corrections near one count require special care because rounding, update timing, noise, and deadtime may dominate.

Quantity to inspect	Why it matters
PWM source and timer clocks	Determine tick duration and available period counts.
Count mode and period register	Determine actual duty levels and resolution.
Compare registers	Show the quantized commands applied to each channel.
Deadtime registers	Show the blank interval inserted between complementary edges.
HRPWM configuration	Shows whether fractional-tick edge placement is active.
Gate pulse on oscilloscope	Confirms that timer output becomes a practical hardware pulse.
Switch-node waveform	Confirms the electrical action after the driver and devices.

14. Worked Example

Consider center-aligned PWM with a 200 MHz time-base clock and a switching frequency of 40 kHz. The period register is approximately 2500 counts, giving an ordinary duty step of 1/2500, or 0.04%. The effective resolution is about 11.29 bits. If the DC input is 800 V, one ideal average-voltage step is 0.32 V.

$$N = \frac{200,000,000}{2 \times 40,000} = 2500$$

$$\Delta D = \frac{1}{2500} = 0.0004 = 0.04\%$$

$$\Delta V = 800 \times 0.0004 = 0.32 \text{ V}$$

A requested base duty of 10% corresponds to about 250 compare counts and is easy to represent. A compensation of 0.25 V, however, corresponds to only 0.03125% duty, or 0.78125 count. It cannot be represented exactly by ordinary PWM. Depending on the initial fractional position and rounding method, it may produce either no count change or one count. The base duty is feasible while the fine compensation is resolution-limited.

If switching frequency is lowered to 20 kHz, the period register becomes approximately 5000 counts, the duty step becomes 0.02%, and the voltage step becomes 0.16 V. The same 0.25 V correction now corresponds to 1.5625 counts and is easier to distinguish. This improvement comes at the cost of increased inductor-current ripple and changed sampling behavior.

Case	Period count	Duty step	Voltage step at 800 V
40 kHz center-aligned	2500	0.04%	0.32 V
20 kHz center-aligned	5000	0.02%	0.16 V
10 kHz center-aligned	10000	0.01%	0.08 V

15. Verification Procedure

The most reliable verification separates the controller from the PWM peripheral and then separates the PWM peripheral from the power stage. First, disable closed-loop correction and command a fixed base duty. Record the floating-point duty, period register, and compare register. Increase duty in small known increments and identify the smallest increment that changes the compare register. This directly measures digital quantization.

Second, measure the PWM output with an oscilloscope. Confirm the period, ON time, complementary relationship, and deadtime. Reduce the pulse command until the pulse becomes distorted or disappears;

this identifies practical minimum-pulse behavior. Third, observe the switch node to confirm that the gate pulse produces the expected electrical transition. Finally, restore closed-loop control and compare the controller's requested correction with the measured compare-count changes.

A useful sweep begins with increments larger than one expected count and then reduces them. For example, test duty differences of 0.10%, 0.05%, 0.02%, 0.01%, and 0.005%. The smallest increment that reliably changes both the compare register and measured pulse is the practical command resolution for that configuration.

16. Design Tradeoffs and Improvement Methods

If ordinary duty resolution is insufficient, several approaches are available. Lowering PWM frequency increases the count range but increases ripple. Raising the timer clock improves resolution if the peripheral supports the higher rate. Edge-aligned operation may provide more compare levels than center-aligned operation at the same frequency, though it changes waveform symmetry and sampling options. High-resolution PWM can provide fractional-tick control with additional configuration complexity.

Control design can also account for quantization. A small deadband can prevent unnecessary switching between adjacent levels when the remaining error is acceptable. Conditional integration can reduce limit cycles and windup. Dithering or pulse-density methods can alternate between adjacent counts to create the required long-term average, at the cost of added ripple. Increasing actuator sensitivity may make one duty count produce a larger correction, but that can reduce fine control rather than improve it. The best method depends on the accuracy requirement, plant filtering, noise, losses, and acceptable ripple spectrum.

17. Summary

Duty cycle is a ratio of ON time to switching period, but digital PWM duty is not continuously variable. The timer clock and carrier mode create a finite set of edge positions. The configured period register, not merely the compare-register bit width, determines ordinary effective resolution. Higher PWM frequency reduces the number of timer counts available per period, while lower frequency increases them. The electrical meaning of one duty count also grows with DC-link voltage.

The most important practical distinction is between requested duty, quantized timer duty, and realized hardware pulse. A controller can calculate corrections that are smaller than one timer count. Proportional action may then leave a residual error, while integral action may move repeatedly between adjacent levels and create a limit cycle. Deadtime, minimum pulse width, driver delay, semiconductor behavior, and simulation time step can make practical resolution worse than timer mathematics predicts. Reliable design therefore requires calculation from the actual PWM configuration, inspection of compare registers, and measurement of gate and switch-node waveforms.

18. References

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